REWEAR-FUSED: Intellectual-Property Landscape, Patent White-Space & Freedom-to-Operate Analysis (current as of June 2026)

TL;DR

- **The single most defensible IP position is the matched-pair co-design itself**: a *system and method* claim covering the simultaneous creation of a purpose-built enzymatically-cleavable aliphatic segmented PU-urea elastane fiber AND a de-novo urethanase designed against that fiber's engineered cleavage site. This "design the lock and the key together" approach appears to be genuine white space — every major incumbent (Carbios, Samsara Eco, Protein Evolution, Epoch Biodesign, Covestro/Greifswald) does the opposite: engineering enzymes to attack pre-existing commodity plastics.
- **Freedom-to-operate is favorable but not unconditional.** The dangerous patents are composition-of-matter claims on specific enzyme sequences (Covestro WO2023194440A1 / US 12,351,853; the EP3587570A1 urethanase family) and Carbios's process claims (US 10,124,512). REWEAR-FUSED can design around all of them because its enzyme is de-novo (low sequence identity to UMG-SP2 and to claimed SEQ IDs) and its fiber is novel — but the team must keep enzyme sequence identity low and avoid the specific two-step glycolysis-then-urethanase process language.
- **"Active progress toward securing IP" for Cox is achievable now** by filing one or two provisional patents (USPTO filing fee as low as $60 micro-entity / $130 small-entity) before the residency, anchored on the co-design method and the specific fiber+enzyme pair, plus a defensive-publication strategy for the Digital Recyclability Passport layer.

Key Findings

1. **The enzymatic-recycling field is overwhelmingly "enzyme-to-existing-plastic."** Carbios holds ~336 patent titles in 53 families (end-2022); Samsara Eco, Protein Evolution, and Epoch Biodesign all design enzymes to degrade commodity PET, nylon, and polyurethane that were never designed for recyclability. None co-designs a novel polymer with its enzyme.
2. **Enzymatic polyurethane/elastane degradation is early-stage and the key IP is narrow (sequence-specific).** The foundational urethanase patents (Greifswald/Covestro) claim *specific sequences* (SEQ ID NO:1–3 and variants within 15% identity), not the concept of urethane-bond cleavage. This leaves wide room for a de-novo enzyme.
3. **De-novo enzymes are patentable as compositions of matter** if the application discloses the actual sequence and demonstrated function; the open-source design tools (RFdiffusion2, LigandMPNN, Boltz-2, ProteinMPNN, FoldSeek) carry permissive licenses (MIT/open) that do NOT encumber ownership of the designed output.
4. **Recyclable/degradable elastane is a crowded but shallow field**: Hyosung (bio-BDO regen), LYCRA (EcoMade/QIRA), and Asahi Kasei (ROICA V550 partly-degradable, ROICA EF

recycled) all pursue bio-based or mechanically-recycled spandex — none builds an *enzymatically-cleavable bond into the soft segment* matched to a bespoke enzyme.

5. **The matched-pair co-design is, to the best available evidence, novel.** The closest prior art (US 6,946,284 photolyase + designed cleavable polymer; Ting Xu/Berkeley embedded-enzyme PLA/PCL) uses *natural* enzymes, not de-novo enzymes co-designed with a novel fiber.

Details

PART 1 — The Enzymatic Plastic/Textile Recycling Patent Landscape

(a) Carbios (Clermont-Ferrand, France). Carbios is the global leader in enzymatic PET recycling. As of end-2022 it held **336 patent titles in 53 patent families** (up from 38 families / 33 grants at end-2020), spanning Europe (40 titles), North America (41 US + 23 Canada), and Asia (152, including 37 China, 27 Japan, 24 India) (Business Wire, 28 Feb 2023; carbios.com). Key families:

- **US 10,124,512 B2** — "Method for recycling plastic products" — filed Nov 2013, protects enzymatic PET depolymerization into monomers; described as protecting the innovation **through 2033** (recyclingportal.eu). This is the foundational process patent. It broadly recites contacting plastic with a degrading enzyme and even lists enzyme classes (cutinase, lipase, esterase, amidase, etc.).
- **US 10,385,183** — "Process of recycling mixed PET plastic articles" — covers depolymerase use with plastic-binding modules and mutated enzymes with improved PET affinity.
- The engineered **LCC-ICCG cutinase variant** (F243I/D238C/S283C/Y127G; Tournier et al., *Nature* 580:216, 2020) achieves, per the paper, "a minimum of 90 per cent PET depolymerization into monomers, with a productivity of 16.7 grams of terephthalate per litre per hour (200 grams per kilogram)" over 10 hours; protected by enzyme-composition patents (the "PET enzymes described in *Nature*" granted in the US).
- Carbios also has **FR3088070A1** (enzymatic PET degradation near the polymer Tg) and the Carbiolice "Evanesto"/masterbatch family (e.g., US 11,773,257 "biodegradable polyester article comprising enzymes"). Carbios stated from 2023 it would "focus efforts on protecting innovation related to the enzymatic degradation of OTHER polymers" — a signal it intends to expand toward PU/PA.
- **Relevance to REWEAR-FUSED:** Carbios's claims are PET/polyester-centric and process- or sequence-specific. They do not cover a novel PU-urea fiber or a de-novo urethanase.

(b) Samsara Eco (Australia). Uses an AI-driven enzyme platform ("EosEco") to depolymerize PET, nylon 6,6, and nylon 6, including dyed and nylon/elastane-blend textiles. Foundational patent filed ~**September 2021** (polyester recycling — engineered enzymes breaking polymers into monomers); subsequent international (PCT) applications by **August 2022** covering nylon/polyester blend deconstruction (WWD/Sourcing Journal). Samsara also patented a **pre-processing step** that transforms dense plastic into an enzyme-accessible state "in minutes, not days" (Trellis). Backed by Lululemon and Woolworths; opened a commercial plant at Jerrabomberra (near Canberra) in 2025. Sustainable Plastics WWD

- **Relevance:** Samsara processes *elastane-blended* textiles, but as a contaminant/feedstock it separates — it does not make a new recyclable elastane. Its IP is enzyme + pre-processing for existing fibers.

(c) Protein Evolution Inc. (New Haven, CT). "Biopure" technology; AI-designed enzymes for polyester, expanding (via a January 2024 partnership with **Basecamp Research**) to **polyurethane and nylon**. Patent filings (assignee "Protein Evolution Inc.," Justia) include **WO2019168811A1** ("Enzymes for polymer degradation" — modified PETase) and a co-filed family with PSL/CNRS/ESPCI/Sorbonne on **pretreatment + enzymatic degradation of crystallizable polymers**, plus a particle-separation/denaturation process patent. Their website states "All IP retained by you" for client enzyme-engineering work. Protein-evolution

- **Relevance:** Designs novel enzymes for *existing* PU/nylon — the explicit opposite of co-design.

(d) Other players & academic IP.

- **Epoch Biodesign (London).** AI-designed enzymes; "patented enzymatic recycling process" for nylon 6,6 including **elastane-blended textiles** and silicone-coated airbag fabric; raised >$50M (Inditex, Lowercarbon, Extantia); demo plant at Imperial College targeting hundreds of tonnes/yr (2026). Directly relevant competitor on elastane-blend handling, but again enzyme-to-existing-fiber. Specific granted patent numbers were not publicly identified; the company describes its process as "patented."
- **Greifswald / Covestro urethanase patents (the most directly relevant FTO risk).** Bornscheuer group (Branson et al., *Angew. Chem. Int. Ed.* 2023, e202216220) discovered metagenomic urethanases UMG-SP1/2/3 using a **Ser–Ser_cis_–Lys** triad (amidase signature superfamily).
  - **EP3587570A1 / WO2019243293** — "Novel urethanases for the enzymatic decomposition of polyurethanes" (priority ~2018).
  - **WO2023194440A1** (assignee **Covestro Deutschland AG**; priority **8 Apr 2022**, filed 5 Apr 2023; national phase US 18/853,763 → **US 12,351,853**; also EP/CA/CN/JP). Claims a polypeptide of **SEQ ID NO:1, 2 or 3 or a variant within 15% (preferably 10%, 5%)** amino-acid changes having urethanase activity, plus the two-step **glycolysis-then-urethanase** PU recycling process. Earlier WO2006/019095 and WO2013/134801 (EC-3 enzyme on aromatic PU) are also cited.
  - **UMG-SP2 structure** (PDB 8WDW; structure-guided engineering yielding >30-fold higher depolymerization activity, PMC11967865) and the **ACS Catalysis 2025** "urethanase-5" (51% identity to UMG-SP2). 2025–2026 work extended metagenomic

urethanases toward **elastane** specifically, but a 2021 ChemSusChem review notes "no enzymatic system has shown to be effective at biodegrading elastane."

- **NREL/BOTTLE** — WO2022226109A1 ("Polymer degrading enzymes," PET hydrolases, may be active on polyester-polyurethanes; priority 20 Apr 2021).

(e) FTO picture for enzymatic PU/elastane depolymerization. The space is **narrowly claimed and sequence-specific**. There is no broad, dominating patent on "cleaving urethane bonds with an enzyme" per se; the strongest claims are to *named sequences and their close variants* (Covestro SEQ ID 1–3 ±15%) and to *specific processes* (glycolysis→urethanase). A de-novo urethanase with low identity to UMG-SP2/SEQ ID 1–3, used on a novel fiber, sits in open space. The catalytic *mechanism* (Ser-His-Asp or Ser-Ser-Lys triad) is not patentable subject matter — mechanisms/laws of nature cannot be claimed, only specific sequences and processes.

PART 2 — De-Novo / Computational Enzyme-Design IP Landscape

(a) How designed enzymes are protected. In the US and EU, a de-novo protein is patentable as a **composition of matter** if the specification discloses the actual amino-acid sequence (SEQ ID) and a demonstrated/credible function (USPTO MPEP 2163 written-description; the *Amgen v. Sanofi* enablement constraints push toward narrow, sequence-anchored claims). Practice (and the 2008 USPTO written-description guidelines, Example 11) permits genus claims of the form "polypeptide ⩾X% identical to SEQ ID NO:Y having activity Z," but post-*Amgen* and PTAB decisions like *Ex parte Porro* mean broad percent-identity genus claims face written-description/enablement scrutiny; the most defensible claims are the **specific sequence + narrow identity band (e.g., ⩾90–95%) + defined function/assay**. Designed enzymes from the Baker lab are routinely claimed exactly this way (e.g., US 2023/0295639 "De novo designed NTF2-like scaffolds for de novo design of enzymes," claiming SEQ ID NO:1–1615 with percent-identity tiers).

(b) Baker lab / IPD patent estate and spinouts. The Institute for Protein Design (University of Washington) owns the patents; spinouts take licenses. RFdiffusion was released "free for both non-profit and for-profit use under a governed license." Designed-enzyme/serine-hydrolase work (Lauko et al., "Computational design of serine hydrolases," bioRxiv 2024) is covered by a **University of Washington provisional (App. 63/535,404) on the ChemNet network**; D. Baker and I. André are inventors; **Vilya** licensed it. Spinouts: **Vilya** (peptides/macrocycles; exclusive RFpeptides license), **Xaira Therapeutics** ($1B launch, antibodies/protein design), **EvolutionaryScale** (ESM models), Monod Bio, Outpace Bio. Crucially, **using RFdiffusion/LigandMPNN outputs does not put the user under any IPD/Baker patent** — the patents cover specific designed *molecules/networks*, not the act of running the tools, and the tool licenses permit commercial use. bioRxiv

(c) IP status of the open-source tools REWEAR-FUSED uses.

- **ProteinMPNN / LigandMPNN** — code and model parameters under the **MIT license** (dauparas/LigandMPNN GitHub); designs are freely ownable and patentable by the user.

- **RFdiffusion / RFdiffusion2 / RFdiffusionAA** — open-source (RFdiffusion under a permissive/"governed" free license incl. for-profit use); RFdiffusion2 code public (RosettaCommons).
- **Boltz-2** — open structure-prediction model (MIT-licensed lineage; successor to AlphaFold-style architectures).
- **FoldSeek** — open-source (GPL-3.0) structural search.
- **Net effect:** None of these licenses claims ownership of generated sequences/structures, and none imposes copyleft on the *biological molecule* you design. REWEAR-FUSED can patent its de-novo enzyme. (Caveat: confirm each tool's exact license text at filing; some model *weights* carry separate, occasionally non-commercial terms — verify before relying.)

(d) Does anchoring on UMG-SP2 risk infringement? Low risk, provided the design is genuinely de-novo. Patents protect **sequences (and close variants), not catalytic mechanisms or active-site geometries.** "Anchoring on but not copying from" UMG-SP2 — i.e., borrowing the catalytic-triad concept while generating a new backbone/sequence with low identity — avoids the Covestro SEQ ID 1–3 (±15%) claims and the UMG-SP2 sequence claims. The one true hazard is landing within a claimed identity band; keep identity to any claimed urethanase well below the claimed thresholds (target <50%, ideally <40%), document the de-novo generation trajectory, and verify with FoldSeek/BLAST that the final sequence does not read on any granted SEQ ID.

PART 3 — Recyclable/Degradable/Bio-Based Elastane & PU Landscape

(a) Bio-based & recycled spandex.

- **Hyosung (creora/regen).** World's largest spandex maker; first to commercialize bio-based spandex. creora® bio-based spandex initially replaced **30% of petroleum resources with bio-material from industrial "dent" corn** and, per a third-party LCA underpinning its July 2022 SGS certification, "reduces its carbon footprint by 23 percent as compared to the production of regular spandex"; bio-content has since been raised toward 70%. Hyosung is transitioning bio-BDO feedstock from corn to **sugarcane** via partnership with **Geno (Genomatica)** (Vietnam plant). It also offers "regen" recycled spandex (RCS-certified, 100% reclaimed waste) and the **R-SCAN digital product passport** process. These are **bio-/recycled-feedstock** plays, not cleavable-by-design.
- **The LYCRA Company (EcoMade / QIRA).** LYCRA EcoMade uses bio-derived BDO (QIRA, partnership with Qore) and recycled content. Again feedstock substitution, not engineered cleavability.
- **Asahi Kasei (ROICA). ROICA EF** = world's first recycled spandex (>50% recycled, GRS-certified). **ROICA V550** = "world-first partly-degradable" elastane: under ISO 14855-1 (tested by OWS), "around 35 per cent of the yarn breaks down within 270 days under industrial composting conditions, without releasing harmful substances"; it is also a Cradle-to-Cradle Material Health Gold yarn. Notably degradation is *passive/compost* and *partial,* not enzyme-triggered or monomer-recovering.

(b) Cleavable / dynamic-covalent / depolymerizable PU patents. A substantial academic and patent literature exists on **biodegradable polyurethanes with hydrolytically/enzymatically cleavable moieties** in the soft or hard segment:

- **US 6,221,997 B1** "Biodegradable polyurethanes" — segmented PU with **cleavable ester sites in the chain extender, arranged to be enzyme-recognizable** (amino-acid-based chain extenders recognized by enzymes). This is meaningful prior art on enzyme-cleavable segmented PU — but it pairs the polymer with *natural* proteolytic enzymes (chymotrypsin), is aimed at biomedical degradation, and does **not** co-design a bespoke enzyme. Google Patents
- Academic: ester-cleavable soft segments enabling fast glycolysis (J. Polym. Environ. 2025); chemoenzymatic poly(ester-urethane)s recyclable by **lipase** in dilute solvent (PMC3189728); biodegradable poly(ester-urethane)urea elastomers (Acta Biomater.). All use existing enzymes.
- Dynamic-covalent/PDK elastomers (LBNL Helms/Persson) — chemically (acid) recyclable, with authors explicitly "envisioning co-designing future elastomers" for circularity — but via chemical, not enzymatic, recycling.

(c) Enzymatically degradable polyester-polyurethanes specifically. Polyester-PUs are cleavable at ester bonds by cutinases/lipases (long-known); polyether-PUs' urethane bonds were "inaccessible" until the 2023 urethanases. Magnin et al. screened 50 hydrolases, finding an amidase (E4143) and esterase (E3576) active on PU. All are enzyme-to-existing-PU. ScienceDirect

(d) The white space. There is **no identified patent on a NEW enzymatically-cleavable aliphatic segmented PU-urea elastane with the cleavable bond engineered into the soft segment AND matched to a de-novo enzyme.** US 6,221,997 is the nearest art but (i) targets biomedical resorption, (ii) relies on natural proteases, (iii) is not an elastane fiber product, and (iv) does not co-design the enzyme. REWEAR-FUSED's specific composition — an aliphatic (non-aromatic-isocyanate) segmented PU-urea spandex with a defined enzyme-labile linkage placed in the soft segment for near-neutral-pH enzymatic depolymerization to recoverable monomers — appears to be open.

PART 4 — Matched-Pair Co-Design Novelty & White-Space Argument

(a) Prior art on co-designing polymer + matched enzyme together. Based on a dedicated prior-art sweep, the *simultaneous* de-novo co-design of a novel polymer AND a bespoke enzyme built for it is **genuine white space**. The nearest neighbors:

- **US 6,946,284** "Solubilizing cross-linked polymers with photolyase" — a polymer is *deliberately synthesized* with [2+2]-cyclobutane motifs matched to a *natural* photolyase. Conceptually the closest "lock-and-key" patent, but the enzyme is natural and the field is crosslinked-polymer solubilization.
- **Ting Xu / UC Berkeley embedded-enzyme plastics** (DelRe et al., *Nature* 2021; US 12,344,880 "Mechanoenzymatic degradation of polymers"; US 12,559,604 "Biodegradable

polymer-additive blends") — "programmed degradation" pairing enzyme+polymer, but using **commercial natural enzymes (proteinase K, lipase) in commodity PLA/PCL**. Their own patent language ("there are numerous opportunities to program plastic lifecycle via rational design of enzyme-embedded plastics") frames the design-pairing as forward-looking.

- All incumbents (Carbios, Samsara, Protein Evolution, Epoch, Covestro/Greifswald) are explicitly **enzyme-to-existing-plastic** — the opposite paradigm.
- Reviews (ACS Synth. Biol. 2024, PMC11264326, "how future polymers can be designed to further enhance their biodegradation"; Chem Soc Rev 2023, D3CS00556A) treat polymer-for-enzyme co-design as an **aspirational/future direction**, supporting novelty. No source explicitly negates the concept — its near-total absence from the literature is itself the white-space signal.

(b) Strongest novelty argument & what is patentable. REWEAR-FUSED can seek protection on four distinct fronts:

1. **The co-design METHOD** — a computational pipeline that jointly optimizes a cleavable-fiber chemistry and a de-novo enzyme against the engineered cleavage site (system/method claim). *Most novel; hardest to invent around.*
2. **The matched fiber+enzyme SYSTEM** — a composition/kit comprising the specific cleavable PU-urea fiber AND the cognate de-novo enzyme. *Strong, commercially central.*
3. **The fiber COMPOSITION** — the specific aliphatic segmented PU-urea with the defined soft-segment cleavable linkage.
4. **The enzyme COMPOSITION** — the de-novo urethanase sequence(s) (SEQ ID + identity band + activity).

The **most defensible** is the combination of (2) the matched system and (1) the co-design method: even if a competitor independently makes a cleavable elastane OR a urethanase, the *paired, mutually-optimized* system and the method that produces it remain protected.

(c) Likely attorney-drafted claims. Expect: (i) independent **system/composition** claim — "A recyclable textile system comprising (a) a segmented polyurethane-urea elastane fiber having an enzymatically-cleavable linkage in its soft segment, and (b) a non-naturally-occurring polypeptide having carbamate-hydrolase activity at pH 6–8 that catalyzes cleavage of said linkage"; (ii) **fiber composition** claims (independent of enzyme); (iii) **enzyme** claims (SEQ ID + ⩾X% identity + activity assay); (iv) **method-of-recycling** claims (contacting the fiber with the enzyme to recover monomers); (v) **method-of-co-design** claims. The **strongest/broadest defensible** independent claim is the system claim, because it captures the commercial product and the core inventive concept while being anchored to concrete, enabled structures.

(d) Obviousness risk and rebuttal. A challenger will argue: cleavable PUs are known (US 6,221,997), de-novo enzyme design is known (Baker), and combining them is obvious. **Non-obviousness rebuttals:** (i) no prior art teaches *co-optimizing* both halves against each other — the prior arts each modify only one side; (ii) **teaching away** — the entire industry invests in attacking *existing* recalcitrant plastics, implying the field did not consider redesigning the substrate as the solution; (iii) unexpected results — a matched pair can achieve near-neutral-pH, near-complete, monomer-recovering depolymerization that neither known cleavable PUs (partial/compost, e.g., ROICA's ~35% in 270 days) nor known urethanases (slow, low-yield on real PU) achieve; (iv) long-felt need — elastane recyclability is explicitly unsolved ("no enzymatic system effective at biodegrading elastane," 2021). Keep contemporaneous records of design choices and performance data to support secondary indicia.

PART 5 — Practical IP Strategy & FTO Conclusion

(a) Provisional filing strategy (12-month path).

- **File first (Provisional #1, immediately):** the **co-design method + matched fiber+enzyme system**, with as much enabling detail as exists (fiber chemistry, the engineered cleavage motif, the de-novo enzyme design protocol, any sequences and assay data). The USPTO provisional *filing fee* is as low as **$60 (micro-entity) / $130 (small-entity)** (37 CFR 1.16(d), fee schedule effective Jan 19, 2025); attorney preparation typically adds $500–$1,500 for a basic filing and $3,000–$5,000+ for a detail-heavy one. A provisional needs no formal claims — but include claim-like "summary of invention" statements and any SEQ listings you have.
- **Provisional #2 (as data matures):** the specific **fiber composition** and **enzyme sequence(s)** once experimentally validated, to anchor written description/enablement.
- **Within 12 months:** convert to a **non-provisional + PCT** (PCT preserves global rights to ~30 months). Prioritize claim scope: system → method → fiber → enzyme.
- **Inventorship/ownership:** secure written assignments from all contributors; confirm no university/employer or hackathon-sponsor IP claims attach; document that tool licenses (MIT/open) leave outputs owned by the team.

(b) FTO conclusion + design-around guidance. REWEAR-FUSED can operate without infringing the major players' patents, subject to three concrete design-arounds:

1. **Enzyme sequence identity** — keep the de-novo urethanase's identity to UMG-SP2 and to Covestro SEQ ID NO:1–3 (and any granted urethanase SEQ IDs) **well below claimed variant bands** (claims reach ±15% changes, i.e., ~85% identity; target <50% identity). Run FoldSeek/BLAST clearance pre-filing.
2. **Process language** — avoid practicing the specific claimed **glycolysis-then-urethanase two-step** process (Covestro) and Carbios's PET-specific process claims; REWEAR-FUSED's direct near-neutral-pH enzymatic cleavage of a designed bond is a different process.
3. **Fiber composition** — the novel aliphatic PU-urea with an engineered soft-segment linkage sits outside commodity-elastane and bio-feedstock claims (Hyosung/LYCRA/Asahi); ensure the chosen cleavable chemistry is not separately claimed (e.g., review US 6,221,997's amino-acid chain-extender claims and design a distinct linkage). Residual risk areas to monitor:

Carbios's stated 2023+ expansion into "other polymers" (PU/PA); Samsara/Protein Evolution/Epoch continuation filings; any Berkeley/Ting Xu de-novo-enzyme continuations.

(c) Credibly claiming "active progress toward securing IP" for Cox. Concrete, truthful statements the team can make after acting: (i) "We have filed a US provisional patent application [number/date] covering our matched-pair co-design method and fiber+enzyme system"; (ii) "We have completed an FTO review identifying the relevant Carbios, Covestro, Samsara, and Protein Evolution families and confirmed design-around paths"; (iii) "Our design tools (RFdiffusion2, LigandMPNN, Boltz-2) carry permissive licenses that leave all designed outputs ownable by us." Even a single \$60–\$130 provisional converts "an idea" into "active IP progress."

(d) Defensive publication vs. patent. Patent the high-value, hard-to-reverse-engineer assets (co-design method, fiber composition, enzyme sequence, matched system). **Defensively publish** (e.g., timestamped disclosure) the parts that are (i) easily independently derived, (ii) not core moat, or (iii) standards-driven — most notably the **Digital Recyclability Passport** compliance layer, which largely implements EU ESPR/DPP requirements (Regulation (EU) 2024/1781, in force 18 July 2024; textile delegated act expected ~2027, mandatory compliance ~2028–2029) and is unlikely to be strongly patentable as it tracks a regulatory standard. Defensive publication blocks others from patenting it while avoiding wasted prosecution cost.

(e) Presenting the IP moat to investors and Cox judges. Frame three layers: (1) **Paradigm moat** — "We design the lock and the key together; incumbents only make keys for old locks. That's a defensible category, not a feature." (2) **Filed-IP moat** — a provisional/PCT on the system + method, with the enzyme and fiber as composition backstops; emphasize that the matched-pair system claim is hard to invent around. (3) **Freedom-to-operate** — a clean FTO story (de-novo enzyme below all claimed identity bands; novel fiber; distinct process) means the company can commercialize without licensing incumbents, while incumbents may eventually need *us* for designed-cleavable fibers. Pair this with the regulatory tailwind (ESPR/DPP mandating recyclability data) to show timing.

Recommendations

1. **Immediately (pre-Cox):** File Provisional #1 (co-design method + matched system). USPTO filing fee is trivial (\$60–\$130 micro/small entity). This alone satisfies "active progress toward securing IP."
2. **Within 30–60 days:** Complete a documented FTO memo against the named families (US 10,124,512; WO2023194440 / US 12,351,853; EP3587570; Samsara/Protein Evolution PCTs); run BLAST/FoldSeek identity clearance on the candidate enzyme.
3. **As data matures (≤12 months):** File Provisional #2 (validated fiber + enzyme sequences); then convert to non-provisional + PCT.
4. **Defensively publish** the Digital Recyclability Passport layer; do not spend prosecution budget there.

5. **Benchmarks that change the strategy:** If a competitor (esp. Carbios, Epoch, or Protein Evolution) publishes/files a *designed-cleavable-fiber* claim, accelerate the fiber-composition non-provisional and narrow enzyme claims. If enzyme identity to any granted urethanase exceeds ~50%, redesign before filing. If ESPR's textile delegated act mandates specific recyclability proof, pivot the Passport from "feature" to "compliance product."

Caveats

- Patent searching here used public web/Google Patents/USPTO/Espacenet snippets; **a professional landscape/FTO search (Derwent, PatBase, full claim charts) is required before relying on white-space and non-infringement conclusions for fundraising or filing.**
- Several incumbent filings (Samsara Eco, Epoch Biodesign, recent Protein Evolution and Carbios continuations) are not fully public or not individually identified by number here; their pending claims could narrow the white space.
- Tool-license terms (especially model *weights* vs. code) occasionally differ and can change; verify each license at filing.
- US enablement/written-description law for sequence genus claims is evolving post-*Amgen v. Sanofi*; claim scope on the enzyme should be drafted conservatively.
- The matched-pair novelty argument is strong but ultimately depends on the specific claims drafted and on art a professional searcher may surface.

Source appendix (key working URLs)

- Carbios portfolio: https://www.businesswire.com/news/home/20230228006180/en/ ; US 10,124,512 — https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/10124512 ; Tournier 2020 — https://www.nature.com/articles/s41586-020-2149-4
- Samsara Eco: https://wwd.com/sourcing-journal/sustainability/samsara-eco-expands-nylon-materials-collective-1238950426/ ; https://www.textileworld.com/textile-world/fiber-world/2025/02/samsara-eco-commercializing-nylon-66-recycling/
- Protein Evolution / Basecamp: https://www.globenewswire.com/news-release/2024/01/24/2815373/0/en/ ; WO2019168811A1 — https://patents.google.com/patent/WO2019168811A1/en ; Justia assignee — https://patents.justia.com/assignee/protein-evolution-inc
- Epoch Biodesign: https://www.epochbiodesign.com/ ; https://www.eu-startups.com/2026/03/epoch-biodesign-raises-e10-3-million-to-use-ai-and-enzymes-to-recycle-plastic-and-textile-waste-at-commercial-scale/
- Urethanases: WO2023194440A1 — https://patents.google.com/patent/WO2023194440A1/en ; EP3587570A1 — https://patents.google.com/patent/EP3587570A1/en ; UMG-SP2 —

https://pmc.ncbi.nlm.nih.gov/articles/PMC11708778/ ; UMG-SP2 structure — https://pmc.ncbi.nlm.nih.gov/articles/PMC11967865/ ; ACS Catal. 2025 — https://pubs.acs.org/doi/10.1021/acscatal.5c01062 ; Branson 2023 — https://onlinelibrary.wiley.com/doi/abs/10.1002/anie.202216220

- De-novo enzyme IP / tools: IPD software — https://www.ipd.uw.edu/software/ ; RFdiffusion license — https://www.ipd.uw.edu/2023/03/rf-diffusion-now-free-and-open-source/ ; LigandMPNN (MIT) — https://github.com/dauparas/LigandMPNN ; serine hydrolase / ChemNet provisional — https://www.biorxiv.org/content/10.1101/2024.08.29.610411v1 ; NTF2 scaffold patent — https://patents.justia.com/patent/20230295639 ; MPEP 2163 — https://www.uspto.gov/web/offices/pac/mpep/s2163.html
- Elastane/PU: Hyosung SGS — https://blog.hyosungtnc.com/hyosung-creora-bio-based-spandex-receives-first-global-sgs-certification/ ; ROICA V550 — https://www.asahi-kasei.co.jp/fibers/en/roica/specialities/index.html ; US 6,221,997 — https://patents.google.com/patent/US6221997B1/en ; chemoenzymatic poly(ester-urethane) — https://pmc.ncbi.nlm.nih.gov/articles/PMC3189728/
- Co-design/embedded enzymes: US 12,344,880 — https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/12344880 ; ACS Synth Biol review — https://pmc.ncbi.nlm.nih.gov/articles/PMC11264326/
- ESPR/DPP: Regulation (EU) 2024/1781 — https://www.carbonfact.com/blog/policy/digital-product-passport-fashion ; https://esprregistry.com/espr-timeline/